Automated Sputtering-Annealing Controller

Phase 02 explanation and operator guide - package version 0.9.20

Purpose. This document explains the fully integrated Phase 02 workflow: direct COSCON UDP control, manual argon leak-valve confirmations, oven PID automation, live safety monitoring, timing information, data logging and safe shutdown.

1. Executive summary

Phase 02 performs three sputtering-annealing cycles by default. The COSCON web interface is not embedded and is not used. The script communicates directly with COSCON IS at 192.168.236.186 on UDP port 2005. The operator still opens and closes the manual argon leak valve when requested.

The integrated control sequence is based on the verified laboratory test in which UDP activation reached stable measured output near 2250 V and 10 mA, remained stable, and returned to Standby.

The dashboard is designed for operation rather than equipment browsing. It shows only the buttons that are valid for the current prompt, plus live telemetry, cycle progress, the current-step countdown, run elapsed time, and known timed work remaining.

Operator supervision remains mandatory. Software cannot replace the COSCON hardware interlock, physical controls, chamber checks or safe laboratory practice.

2. What is automated and what remains manual

Function	Status	How it is handled
COSCON Degas	Automated	Cycle 1 only. The script sends SwitchToDegas and waits for natural Standby.
Target validation	Automated	ValidateOperateTarget is checked before high voltage is requested.
COSCON Operate	Automated	SwitchToOperate uses 10 mA emission and 2250 V energy.
Output qualification	Automated	Mode, interlock, pressure, measured energy and emission must be valid.
Sputtering timer	Automated	The timer runs only after stable measured output is confirmed.
Return to Standby	Automated	The script requests Standby and verifies the safe state.
Argon leak valve	Manual	The operator opens and closes it when the dashboard requests confirmation.
Oven PID setpoint and hold	Automated	S1 is written and verified; the script waits for PV and runs the anneal hold.

3. Complete workflow

Step	Stage	Main behavior	Time information
1	Preflight	Confirm sputter-gun cable and sputtering electronics; click Start run.	Operator-dependent
2	Degassing	Cycle 1: automatic complete COSCON Degas; wait for natural Standby.	Dashboard shows maximum Degas time left
3	Open valve	Operator opens the argon leak valve and stabilizes pressure near 2e-5 mbar.	Operator-dependent
4	Pressure conditioning	COSCON must remain in Standby while pressure stays inside the permitted window.	Default 60 s countdown
5	COSCON activation	Validate target, request Operate, wait for Operating, then qualify actual output.	Transition and qualification countdowns
6	Sputtering	Continuously supervise pressure, interlock, mode, energy and emission.	Default 20 min countdown
7	Automatic Standby	Request Standby and verify Standby or Off before valve closure.	Up to 25 s countdown
8	Close valve	Operator closes the manual argon leak valve fully.	Operator-dependent
9	Oven ramp	Write anneal target and wait for stable PV readings near the target.	Variable; dashboard shows temperature difference
10	Anneal hold	Hold at target for the configured time.	Default 10 min countdown
11	PID reset	Set the configured reset setpoint and continue to the next cycle.	Automatic

Cycles 2 and 3 skip the Degas stage. They begin with the manual valve-opening prompt while COSCON is already expected to be in Standby.

The estimate called **Known timed work left** includes configured timed stages such as expected Degas, pressure conditioning, sputtering and anneal holds. It excludes manual valve waits and oven-ramp time because those durations cannot be predicted reliably.

4. COSCON communication

COSCON uses plain ASCII UDP commands terminated by one carriage-return byte. The integrated client accepts only a strict command whitelist. It does not expose Reset, SetNetwork, StorePreset or DeletePreset.

Command	Role
GetStatus	Read Mode, Interlock and Details. Mode=Error is parsed as a valid status reply.
GetMonitorValues	Read VEnergy, IFilament and IEmission.
GetTargetValues	Read active energy and emission targets for diagnostics.
GetDiagnosticValues	Read diagnostic voltages, currents and temperatures.
SwitchToDegas	Start the complete Degas cycle.
ValidateOperateTarget	Check 10 mA / 2250 V before activation.
SwitchToOperate	Request the validated sputtering target.

Command	Role
SwitchToStandby	Normal stop after sputtering and first safe-stop action.
SwitchToOff	Fallback safe state when Standby cannot be confirmed, or direct stop during Degas.

5. Acceptance conditions and continuous supervision

The sputtering timer starts only after five consecutive valid measured samples. Mode=Operating alone is not enough.

Signal	Normal requirement	Reaction when invalid
COSCON mode	Operating during sputtering	Abort automatic run and enter safe-stop path.
Interlock	Ok	Immediate safe stop; no automatic reset.
Energy	2250 V +/- 50 V	Stop on collapse or sustained out-of-tolerance readback.
Emission	10 mA +/- 1 mA	Stop on out-of-tolerance readback.
Pressure	1e-5 to 5e-5 mbar for activation/sputtering	Activation is blocked or active sputtering is stopped.
Emergency pressure	Below 1e-4 mbar	Immediate pressure safety stop.

During Degas, very low pressure is acceptable. The normal sputtering pressure window is required only after the operator opens the argon valve and before COSCON activation.

A UDP timeout after a non-idempotent command does not prove that the device did not receive it. The software must not blindly resend Operate or stop commands; it checks the resulting device state instead.

6. Dashboard layout

Dashboard area	Information or control
Header	Run name, cycle number, stage, interlock and workflow progress.
Operator guidance	Current instruction and only the action buttons valid at that moment. Abort is always available.
Time information	Current-step countdown, stage elapsed, run elapsed and known timed work left.
Current-cycle workflow	Completed, active, upcoming and skipped stages with planned timing labels.
COSCON and chamber	Mode, details, interlock, energy, emission, filament current and pressure.
Oven and auxiliary	Oven PV/SV and optional Keysight readback.
Live PID control	Manual S1 setpoint write with readback verification.
System status	Active communication or read errors. Cleared after a fully successful monitor cycle.

7. Operator procedure

Before starting

- Confirm that the chamber is under suitable vacuum and all relevant electronics and cables are correctly connected.
- Close the COSCON web interface and SpecsLab/Prodigy so that only Phase 02 sends control commands.
- Keep the physical COSCON controls accessible and remain at the equipment.
- Confirm that COM6, COM9 and COM17 are not being used by another program.

During each cycle

- Follow the dashboard instruction. The only routine manual buttons are the preflight answers, Start run, Valve opened and Valve closed.
- Do not click Valve opened until argon pressure is stable near the recipe target.
- Do not close the leak valve until the dashboard confirms COSCON has returned to Standby.
- Watch the physical COSCON display and chamber behavior even when the software reports healthy telemetry.

Normal finish

- After the final anneal and PID reset, the stage becomes DONE.
- The script saves telemetry, pressure and oven-temperature plots, closes devices and releases serial ports.

Abort / Safe Stop

- Press Abort / Safe Stop for an operator stop, unexpected chamber behavior or suspected hardware problem.
- During Operating or an uncertain state, the script requests Standby once and verifies the result.
- If Standby or Off cannot be confirmed, it requests Off as fallback.
- During Degas, it requests Off directly because this firmware rejected Standby while Degassing in earlier tests.
- On abort, the script also attempts to set the oven PID to the configured abort reset value, default 0 C.

Do not use automatic Reset to clear a COSCON device fault. Inspect the physical state and follow the equipment owner or manufacturer procedure.

8. Data, parameters and troubleshooting

Saved files

- sputter_anneal_log.csv: timestamped cycle, stage, pressure, oven and auxiliary telemetry.

- automation_parameters.json: complete effective run recipe, including launcher overrides.

- pressure_vs_time.png and oven_pid_temperature_vs_time.png: final plots saved at shutdown.

Default recipe values

Parameter	Default
Cycles	3
Expected Degas	20 min; timeout 25 min
Pressure conditioning	60 s
Sputtering	20 min per cycle
COSCON target	10 mA emission, 2250 V energy
Anneal target	620 C
Anneal hold	10 min per cycle
PID reset after anneal	0 C
PID reset after abort	0 C

Troubleshooting

Symptom	Check
COSCON reports Unknown command GetStatus\r	This indicates the old v0.9.13 literal backslash-r bug. Use package v0.9.20 or
Dashboard is blank or does not open	Install pywebview and Microsoft Edge WebView2 Runtime. The dashboard no longer loads the COSCON webpage.
Pressure unavailable	Check COM6, XGS600 connection and whether another process owns the port. Activation remains blocked.
PID write fails	Check COM9, address 00, cable and read/write frame integrity. The script retries and verifies S1.
COSCON Mode=Error	The reply is parsed correctly and triggers the safe-stop path. Do not automatically reset the device.
Next phase cannot start	Use the launcher port-handoff message to identify a COM port that Windows has not released.

This document describes the packaged runtime script. The preserved original_scripts_backup folder is for recovery and comparison only and is never launched automatically.